

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
LAHONTAN REGION

BOARD ORDER NO. 6-94-59
WDID NO. 6B360115010

UPDATED WASTE DISCHARGE REQUIREMENTS

FOR

SAN BERNARDINO COUNTY CONSOLIDATED SERVICE AREA NO. 82—TRONA
AND PIONEER POINT WASTEWATER TREATMENT FACILITY

San Bernardino County

The California Regional Water Quality Control Board, Lahontan Region (Regional Board) finds:

1. Discharger

The San Bernardino County Service Area No. 82, submitted a complete report of waste discharge for the Trona and Pioneer Point Wastewater Treatment Facility on September 9, 1980. On November 30, 1993, Joel Stribling, Assistant Sanitation Facilities Manager, provided some current information for this update. For the purposes of this Order, San Bernardino County Consolidated Service Area No. 82 is referred to as the "Discharger."

2. Facility

The Trona and Pioneer Point Wastewater Treatment Facility is the facility from which the discharge occurs. For the purposes of this Order, the Trona and Pioneer Point Wastewater Treatment Facility is referred to as the "Facility". The Facility discharges domestic sewage and septic tank sludge (septage).

3. History of Previous Regulation by the Regional Board

The Regional Board most recently established waste discharge requirements for the Facility under Board Order No. 6-80-78, which was adopted on October 9, 1980. Prior to the adoption of Board Order No. 6-80-78, the Discharger was discharging domestic sewage collected from the communities of Trona and Pioneer Point at six different designated disposal sites. The discharges at some of these sites were untreated and/or not confined to areas where public access was properly restricted, and were considered to be a nuisance and a threat to public health. Previous waste discharge requirements (Board Order No. 6-76-126) required that (1) the number of discharge points be reduced to facilitate wastewater reuse, and (2) nuisance conditions be corrected. The Discharger subsequently initiated a project to comply with the requirements, using Clean Water Grant funds as a means of financial assistance. As a result of the project, all of the wastewater collected in Trona and Pioneer Point is now discharged to large septic tanks and conveyed via one common outfall pipeline to the North American Chemical Corporation's (NACC) Searles Lake bed recharge trench in Trona, where the Discharger had traditionally discharged a portion of the collected wastewater. The project also included construction of septage drying beds on the eastern edge of the Searles Lake bed to serve the large septic tanks which treat wastewater from the Trona and Pioneer Point collection systems, and several other septic tank systems in the Searles Valley.

8. Disposal Area

Two separate systems collect approximately 0.08 mgd of domestic wastewater from Pioneer Point and are referred to as the North and South Pioneer Point septic tank treatment facilities. The discharge from the North and South Pioneer Point septic tanks flows to the south via a two-mile gravity interceptor pipeline to Trona, where the effluent, and 0.11 mgd of wastewater from septic tanks in Trona, is combined and discharged from a common outfall. The domestic wastewater (0.19 mgd) discharged to the recharge trench commingles with an estimated 20 mgd flow of industrial wastewater from NACC chemical processing plants, and is conveyed to areas on the Searles Lake bed where public access is restricted.

- A. North Pioneer Point: The design-calculated detention time of the North Pioneer Point system is approximately 34 hours (three septic tanks, 100,000 gallons of total storage capacity) with current flows estimated at 0.053 mgd. The Discharger is currently planning to either replace or augment (depending on the condition of the existing tanks after excavation and inspection) the North Pioneer Point system by installation of a new 77,000 gallon septic tank. The purpose of the tank replacement (augmentation) is to ensure the integrity of the disposal system, not to expand treatment capacity; however, augmentation would provide additional treatment and storage capacity.
- B. South Pioneer Point: The South Pioneer Point system has a design detention time of approximately 31.9 hours (two tanks, 30,000 gallons of storage capacity) with current flows estimated at 0.023 mgd.
- C. Trona: The design detention time of the Trona septic tank system is approximately 19.6 hours (two tanks, 90,000 gallons of total storage capacity). An estimated 0.11 mgd of domestic wastewater is collected and treated by the Trona septic tank facility.

9. Authorized Disposal Site

The NACC recharge trench on the west side of Searles Lake bed at Trona is the only authorized disposal site for wastewater. The sludge drying beds on the east side of Searles Lake bed near Trona, and the Baker and Trona/Argus Landfill, located on First Street, two miles north of the community of Argus are the only authorized disposal sites for sludge. The authorized disposal sites are located on land owned or leased by the NACC, with the exception of the sludge drying site and the Baker and Trona/Argus Landfill which are on county lands administered by San Bernardino County Solid Waste Management Department.

10. Site Geology

The Facility is located on the west side of the Searles Valley, in a unique geologic setting. Searles Lake (elevation 1616 feet mean sea level) is a large playa, or dry lake, whose basin surface is encrusted with salts and minerals, cracked from dessication, and dotted by pools of multi-colored brines discharged from chemical processing. Trona Pinnacles National Monument and Recreation area, just south of the Searles Lake bed, offers evidence of previous geologic activity. The surrounding terrain is mountainous; the Argus Range and Panamint Range rise steeply (± 2000 feet) on the west and east sides of Searles Lake, respectively. The predominant economic activity in the area is mining and mineral extraction (from ground water).

11. Site Hydrology

The Searles Valley generally receives less than 5 inches of precipitation annually. Surface runoff is minimal and most precipitation either percolates or evaporates immediately on reaching the Valley floor. Drainage from Salt Wells Canyon, and other ephemeral drainages, feed Searles Lake during occasional periods of flash flooding.

12. Site Hydrogeology

The principle sources of water supply are springs augmented by extractions of ground water, and supplemented by ground water imported via pipeline from Indian Wells Valley. Depth to ground water beneath the Searles Lake bed (where the disposal sites are located) is estimated to be less than five feet. Ground water beneath Searles Lake is hypersaline and considered a prime mineral resource; over thirty salts and minerals are, or have been, economically extracted by chemical processors. Total dissolved solids concentrations commonly range from 389,000 to 450,000 mg/l in the ground water.

13. Receiving Waters

The receiving waters are the ground waters of the South Lahontan Hydrologic Basin, Searles Hydrologic Unit, Searles Hydrologic Area (Department of Water Resources Hydrologic Unit Number 621.10).

14. South Lahontan Basin Plan

The Regional Board adopted a Water Quality Control Plan for the South Lahontan Basin (Basin Plan) on May 8, 1975 and this Order implements the Basin Plan as amended.

15. Beneficial Uses

The only beneficial use of the ground waters of the South Lahontan Hydrologic Basin, Searles Hydrologic Unit, Searles Hydrologic Area as set forth and defined in the Basin Plan is industrial service supply. The beneficial uses of the surface waters of Searles Lake are industrial service supply, non-contact water recreation, wildlife habitat, and saline water habitat.

16. California Environmental Quality Act Compliance

San Bernardino County Board of Supervisors certified a final Environmental Impact Report in accordance with the California Environmental Quality Act (Public Resources Code, § 21000 et seq.) on August 23, 1977. The County is engaged in the continued operation of an existing Facility, and, as such, is exempt from the requirements of the CEQA.

17. Notification of Interested Parties

The Regional Board has notified the Discharger and interested parties of its intent to update waste discharge requirements for the discharge.

18. Consideration of Public Comments

The Regional Board, in a public meeting, heard and considered all comments pertaining to the discharge.

IT IS HEREBY ORDERED that the Discharger shall comply with the following:

I. DISCHARGE SPECIFICATIONS

A. Effluent/Discharge Limitations

1. The total effluent flow of wastewater during a 24-hour period shall not exceed 0.096 million gallons for the North Pioneer Point system; 0.031 million gallons for the South Pioneer Point system; and 0.12 million gallons for the Trona system.
2. The total combined effluent flow of wastewater during a 24-hour period from all septic systems shall not exceed 0.247 million gallons.

B. Receiving Water Limitations

The discharge of waste shall not cause the presence of the following substances or conditions in ground waters of the South Lahontan Hydrologic Basin, Searles Hydrologic Unit, Searles Hydrologic Area:

1. Toxic substances in concentrations that individually, collectively, or cumulatively cause detrimental physiological responses in human, plants, animals, or aquatic life.
2. Identifiable chlorinated hydrocarbons, organophosphates, carbamates, and other pesticide and herbicide groups, in summations, in excess of the lowest detectable levels.

C. General Requirements and Prohibitions

1. There shall be no discharge, bypass, or diversion of raw or partially treated sewage, sewage sludge, grease, or oils from the collection, transport, treatment, or disposal facilities to adjacent land areas or surface waters.
2. Surface flow or visible discharge of sewage or sewage effluent at, or from, the authorized disposal site to adjacent land areas or surface waters is prohibited.
3. The discharge shall not cause a pollution as defined in Section 13050 of the California Water Code, or a threatened pollution.
4. Neither the treatment nor the discharge shall cause a nuisance as defined in Section 13050 of the California Water Code.
5. The discharge of wastewater except to the authorized disposal site is prohibited.

II. PROVISIONS

A. Rescission of Waste Discharge Requirements

Board Order No. 6-80-78 is hereby rescinded.

B. Standard Provisions

The Discharger shall comply with the "Standard Provisions for Waste Discharge Requirements," dated July 1, 1993, in Attachment "B", which is made part of this Order, with the exception of Provision No. 15.

C. Monitoring and Reporting

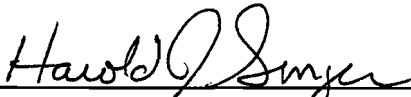
1. Pursuant to the California Water Code Section 13267(b), the Discharger shall comply with the Monitoring and Reporting Program No. 94-59 as specified by the Executive Officer.

2. The Discharger shall comply with the "General Provisions for Monitoring and Reporting," dated July 1, 1993, which is attached to and made part of the Monitoring and Reporting Program, with the exception of Provision No. 1., a-d, f and g.

D. Time Schedule Compliance

The Discharger shall submit a Sludge Management Plan, as described in the attached Monitoring and Reporting Program, by not later than July 15, 1994.

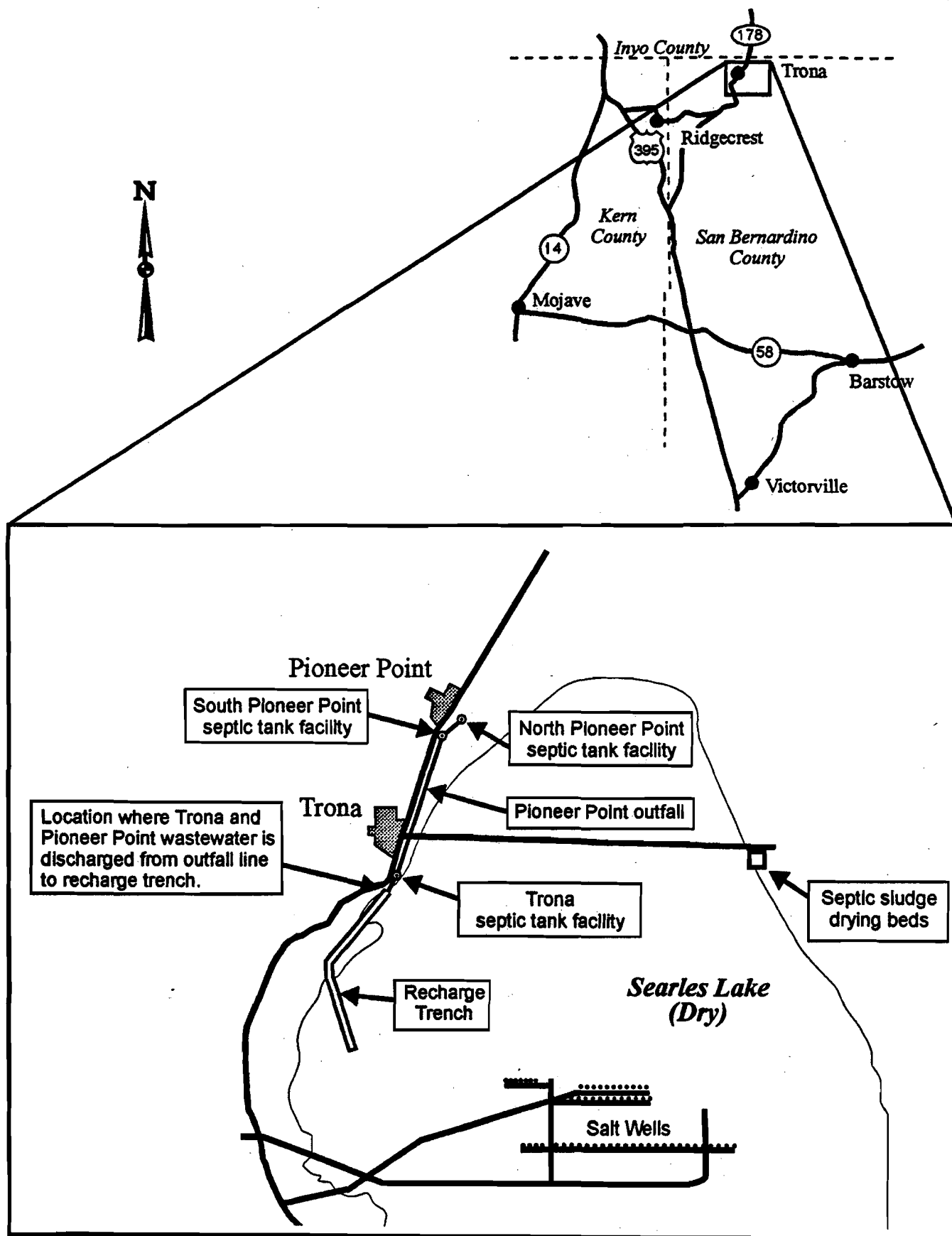
I, Harold J. Singer, Executive Officer, do hereby certify that the foregoing is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, Lahontan Region, on May 19, 1994.



HAROLD J. SINGER
EXECUTIVE OFFICER

- Attachments: A. Location Map
B. Standard Provisions for Waste Discharge Requirements

Attachment "A"
Trona and Pioneer Point
Wastewater Treatment Facility
San Bernardino County



CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
LAHONTAN REGION

MONITORING AND REPORTING PROGRAM NO. 94-58
WDID NO. 6B360115010

FOR

**SAN BERNARDINO COUNTY CONSOLIDATED SERVICE AREA NO. 82—TRONA
AND PIONEER POINT WASTEWATER TREATMENT FACILITY**

San Bernardino County

I. MONITORING

A. Flow Monitoring

The Discharger shall monitor the following:

1. Average monthly potable water usage, in mgd, for each community served by the disposal facilities.
2. The estimated average flow rate, in million gallons per day (mgd), of wastewater to each treatment facility computed for each month.
3. Estimated volume, in cubic yards, of septic tank sludge stored in each septic tank, and remaining septic sludge storage volume available in each septic tank on an annual basis.
4. General condition of collection, treatment, storage, and disposal facilities and immediately surrounding land areas, on a monthly basis, with emphasis on detecting leaks, blockages, surfacing effluent, unusual or excessive vegetation growth, and any other operational problems.
5. Visual inspection of the recharge trench in the vicinity of the septic tank effluent outfall, on a monthly basis, with emphasis on detecting floating debris, and grease and oil (as evidenced by a visible sheen on the water surface) attributable to the septic tank effluent.

B. Plant Effluent Monitoring

Due to the characteristics of the waste being disposed, the unique nature of the disposal area, and the disposal system, monitoring of the septic tank effluent is not required by the Regional Board at this time.

C. Ground Water Monitoring

Due to the characteristics of the waste being disposed, the unique nature of the disposal area, and the single beneficial use designation, monitoring of the ground water is not required by the Regional Board at this time.

D. Sludge Monitoring

1. The Discharger shall submit a Sludge Management Plan by July 15, 1994.
2. The Discharger shall provide an explanation of any deviations from the Sludge Management Plan.
3. Total quantity of septic tank sludge generated (cleaned from any tanks) during the monitoring period.
4. Date and quantity of any sludge landfilled or moved offsite, location of use or disposal, recipient (name and address), and sludge disposal method for all sludge moved offsite.
5. Cumulative total quantity of sludge currently stockpiled on site, including the quantity of sludge added to the stockpile during this monitoring period.

II. REPORTING

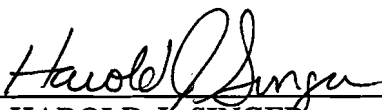
A. General Provisions

The Discharger shall comply with the "General Provisions for Monitoring and Reporting," dated July 1, 1993, which is attached to and made part of this Monitoring and Reporting Program, with the exception of the Provision No. 1, a-d, f and g.

B. Submittal Periods

Beginning on July 15, 1994, a monitoring report including the preceding information shall be submitted to the Regional Board. Subsequent monitoring reports shall be submitted to the Regional Board every six months on January 15 and July 15.

Ordered by:


HAROLD J. SINGER
EXECUTIVE OFFICER

Dated:

May 19, 1994

Attachment: General Provisions for Monitoring and Reporting